Coast vs. Inland: Two Decades of Temperature in Hoquiam and Spokane, Washington

Temitope Dada
CYBR 515 Research Methods & Colloquium

November 3, 2026

1 Introduction

I once visited someone in Hoquiam, a small city on Grays Harbor along Washington's Pacific coast, and the weather there felt nothing like Spokane. Both cities sit in the same state and at almost the same latitude, so I wanted to know whether the data agrees with that impression, and if it does, what exactly makes them different.

Hoquiam faces the open Pacific Ocean. Water heats up and cools down slowly, so a coastal town tends to stay mild all year. Spokane is about 300 miles inland, on the dry eastern side of the Cascade Mountains, where the air can warm quickly under a clear sky and lose that heat just as quickly at night. In this report I compare twenty years of hourly temperature for the two cities. The goal is to show that distance from the ocean shapes not only how hot or cold a place gets, but also how much its temperature swings within a single day.

2 Data

The data comes from the Open-Meteo Historical Weather API [1]. I requested hourly air temperature measured two meters above the ground, in degrees Fahrenheit, for twenty full calendar years: January 1, 2006 through December 31, 2025. The two locations were Hoquiam (46.98 N, 123.89 W) and Spokane (47.66 N, 117.43 W). I asked for times in the local Pacific time zone so that each "day" runs from midnight to midnight local time.

This gives about 175,000 hourly readings for each city. It is worth noting that Open-Meteo's historical data is reanalysis, which means it is a reconstruction built by combining weather models with real observations [2]. It describes the weather over a grid square of several kilometers rather than at one specific weather station, so individual values may differ slightly from official station records.

I downloaded the data with the Python `requests` library, handled it with `pandas`, and drew the figure with `matplotlib`.

3 Methods

Let $T_{d,h}$ be the temperature in hour h of day d, and let n_d be the number of hourly readings in that day. Normally $n_d = 24$, with 23 or 25 on the two days each year when clocks change for daylight saving time.

Raw data points. To keep the plot readable, I show the raw readings every three hours, at $h \in \{0, 3, 6, 9, 12, 15, 18, 21\}$. This still gives eight points per day.

Daily mean. For each day, I used all hourly readings to compute the mean:

$$\bar{T}_d = \frac{1}{n_d} \sum_{h=1}^{n_d} T_{d,h} \tag{1}$$

Daily standard deviation. The spread of temperature within each day is measured with the sample standard deviation, which is the default in `pandas`:

$$s_d = \sqrt{\frac{1}{n_d - 1} \sum_{h=1}^{n_d} \left(T_{d,h} - \bar{T}_d\right)^2} \tag{2}$$

A large s_d means the temperature moved a lot during that day, usually because of a big gap between the afternoon high and the overnight low. On the plot, the shaded band for each day runs from $\bar{T}_d - s_d$ to $\bar{T}_d + s_d$.

Summary measures. To compare the two cities over the whole period, I averaged the daily values across all $D = 7305$ days:

$$\bar{T} = \frac{1}{D} \sum_{d=1}^{D} \bar{T}_d, \qquad \bar{s} = \frac{1}{D} \sum_{d=1}^{D} s_d \tag{3}$$

I computed the same average of s_d for only the days in July and only the days in January to see how the daily swing changes with the seasons.

Colors. The two cities are drawn in colors taken from the viridis palette, dark blue for Hoquiam and green for Spokane. Viridis was designed to stay readable for people with color vision deficiency and to print well in grayscale. I avoided rainbow color schemes, which distort how readers perceive data and are hard to read for people with color vision deficiency [3].

4 Results

Figure 1 shows the full result. Panel (a) covers all twenty years, and panel (b) zooms in on 2021 so the individual readings and daily bands are easier to see. Table 1 lists the summary numbers.

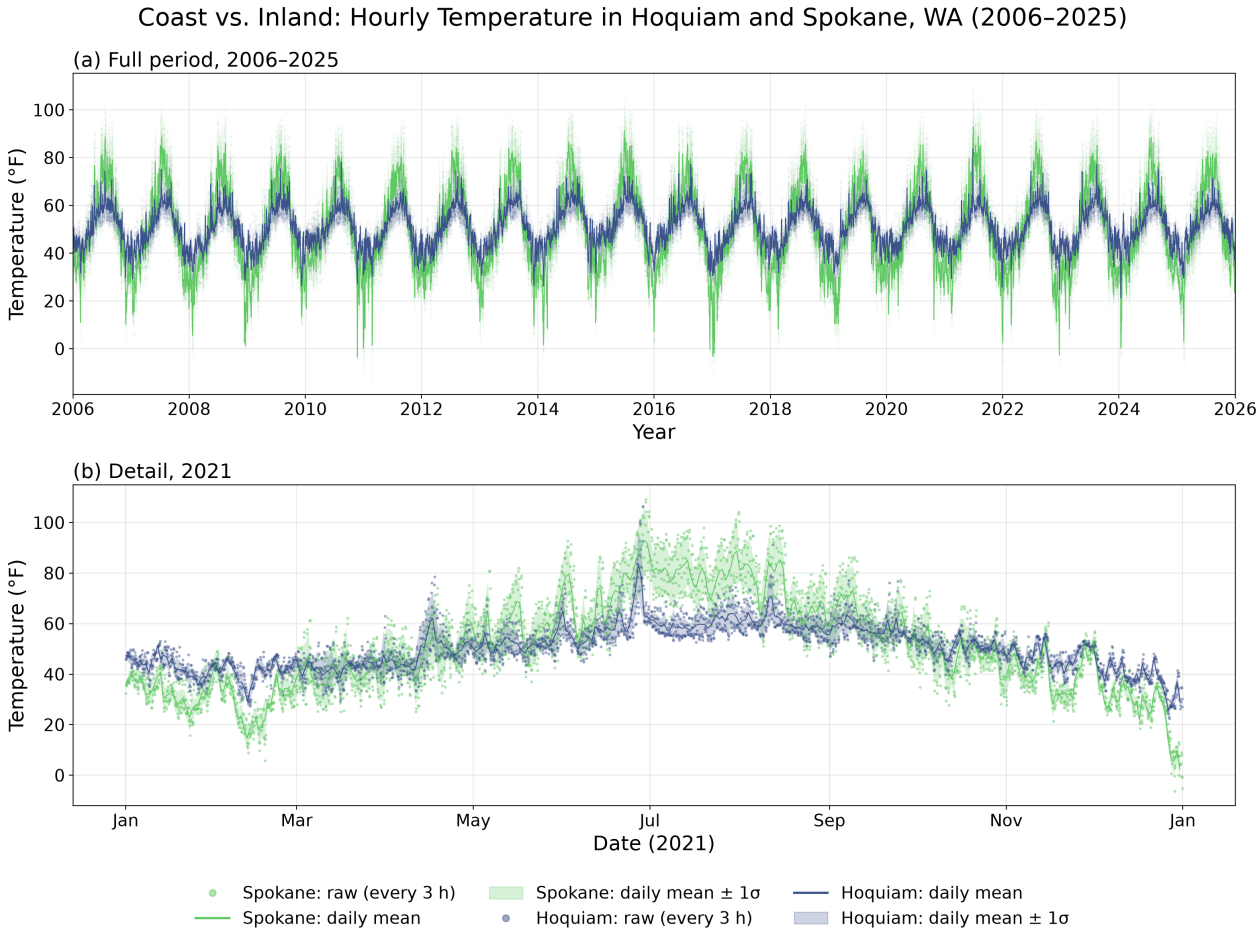


Figure 1: Hourly temperature in Hoquiam (dark blue) and Spokane (green), 2006 through 2025. Faint dots are raw readings taken every three hours, solid lines are daily means, and shaded bands show one standard deviation above and below each daily mean. Panel (a) shows all twenty years. Panel (b) shows 2021 in detail, including the late June heat dome.

Measure	Hoquiam	Spokane
Overall mean temperature, $\bar{T}$	50.7 °F	49.8 °F
Average daily standard deviation, $\bar{s}$	3.9 °F	6.5 °F
Average daily standard deviation, July	5.0 °F	9.0 °F
Average daily standard deviation, January	2.9 °F	4.0 °F
Hottest day (daily mean)	83.6 °F, June 27, 2021	93.0 °F, June 29, 2021
Coldest day (daily mean)	21.2 °F, Jan. 13, 2024	−3.6 °F, Nov. 24, 2010
Highest raw reading	106.4 °F, June 28, 2021	109.2 °F, June 29, 2021

Table 1: Summary of the twenty year record, computed with Equations (1) to (3).

4.1 Same average, very different years

The most surprising number is the first one. Over twenty years, the two cities have almost the same average temperature, 50.7 °F in Hoquiam and 49.8 °F in Spokane, less than one degree apart.

If you only looked at the average, you would think they have the same climate.

Figure 1(a) shows why that would be wrong. Every year, Spokane's green line climbs well above Hoquiam's in summer and drops well below it in winter, often falling close to 0 °F during cold snaps. Hoquiam's dark blue line stays inside a much narrower range, mostly between about 30 and 70 °F. Spokane gets its average by balancing hot summers against cold winters. Hoquiam gets the same average by simply staying mild.

4.2 The daily swing

The standard deviation bands tell the second half of the story. On an average day, Spokane's temperature varies about two thirds more than Hoquiam's (6.5 °F against 3.9 °F). The difference is largest in July, when Spokane's daily standard deviation is 9.0 °F, almost twice Hoquiam's 5.0 °F. You can see this in Figure 1(b), where the green band is much wider than the blue one through the summer months.

This matches what we would expect. Summer days in Spokane are usually dry and clear, so the sun heats the ground quickly in the afternoon, and that heat escapes into the sky at night. In Hoquiam, humid ocean air and frequent clouds act like a blanket, so nights stay closer to the daytime temperature. In January the gap is smaller (4.0 °F against 2.9 °F), because both cities spend much of the winter under clouds.

4.3 A closer look at 2021

Figure 1(b) shows how the two cities behave across one year. In winter Spokane is clearly colder, including a cold snap in February when its daily mean fell to around 15 °F while Hoquiam stayed near 30 °F. In spring and again in October, the two lines meet and overlap, and for a few weeks the cities have almost the same weather.

The most dramatic part of the year is the heat dome at the end of June 2021. Both cities had their hottest day of the entire twenty years during this event. Hoquiam peaked first, with a daily mean of 83.6 °F on June 27 and a raw reading of 106.4 °F at noon on June 28. Spokane peaked two days later, on June 29, with a daily mean of 93.0 °F and a reading of 109.2 °F at 3 p.m. The two day delay suggests the heat built on the coast first and then moved inland.

What happened next is just as telling. Within a few days, Hoquiam dropped back to around 60 °F as cool ocean air returned, and it stayed there for the rest of the summer. Spokane did not have an ocean to cool it down and its daily mean stayed between about 75 and 90 °F through much of July and August. The heat dome was the one time the ocean failed to protect Hoquiam, most likely because winds blew from the land toward the sea and blocked the usual cool air. The year then ended with the opposite extreme: in late December, Spokane fell to around 0 °F while Hoquiam stayed in the mid 20s and above.

5 Conclusion

Over 20 years, Hoquiam and Spokane share almost the same average temperature, yet they live in very different weather. Spokane has hotter summers, colder winters, and a daily swing that is nearly twice as large in July. Hoquiam stays mild because the Pacific Ocean holds its temperature steady. The one clear exception was the 2021 heat dome, which pushed both cities to their hottest days on record in this dataset. The data confirm what I felt on that visit: the coast and the inland are two different worlds, even inside one state.

References

[1] P. Zippenfenig, *Open-Meteo.com Weather API*, Zenodo, 2023. Historical Weather API: `https://open-meteo.com/en/docs/historical-weather-api`

[2] H. Hersbach et al., "The ERA5 global reanalysis," *Quarterly Journal of the Royal Meteorological Society*, vol. 146, no. 730, pp. 1999 to 2049, 2020.

[3] F. Crameri, G. E. Shephard, and P. J. Heron, "The misuse of colour in science communication," *Nature Communications*, vol. 11, article 5444, 2020. `https://doi.org/10.1038/s41467-020-19160-7`